



EBN 122

Chapter 5: Operational Amplifiers (Op Amps)

Chapter 5 overview

✓ 5.2. Operational Amplifiers (Op Amps)

✓ 5.3. Ideal Op Amps

5.4. Inverting Amplifiers

5.5. Noninverting Amplifiers

5.6. Summing Amplifiers

5.7. Difference Amplifiers

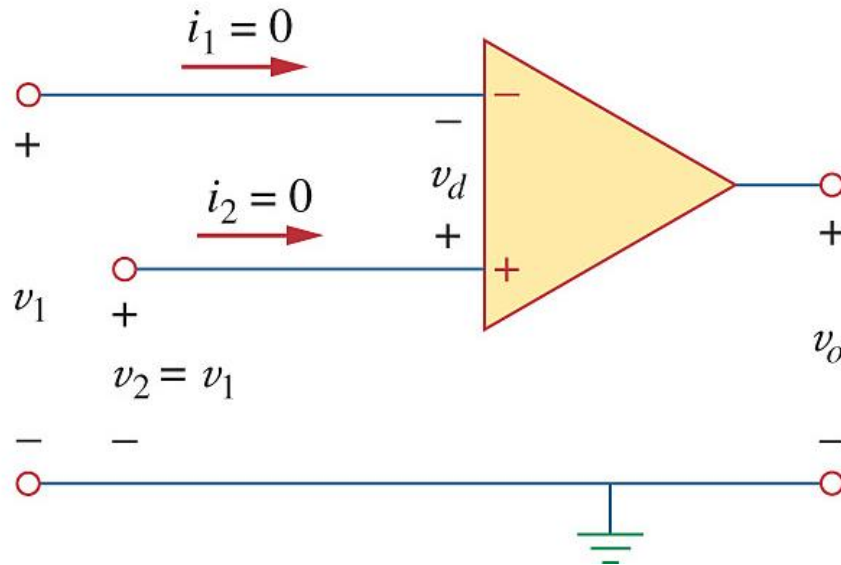
5.8. Cascaded Amplifier Circuits



Video lecture 2

Overview

For ideal op amps:

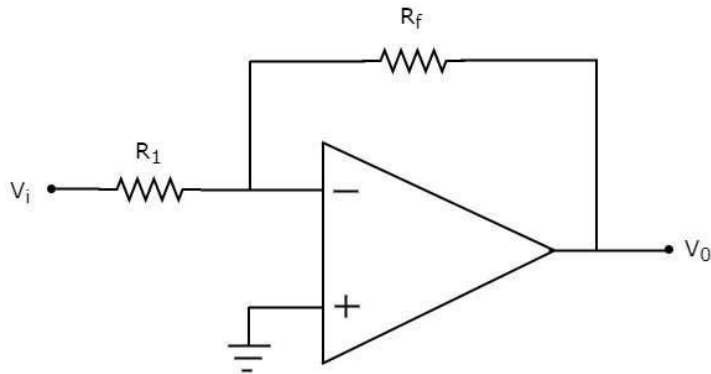


$$\begin{aligned} i_1 &= i_2 = 0 \\ v_1 &= v_2 \end{aligned}$$

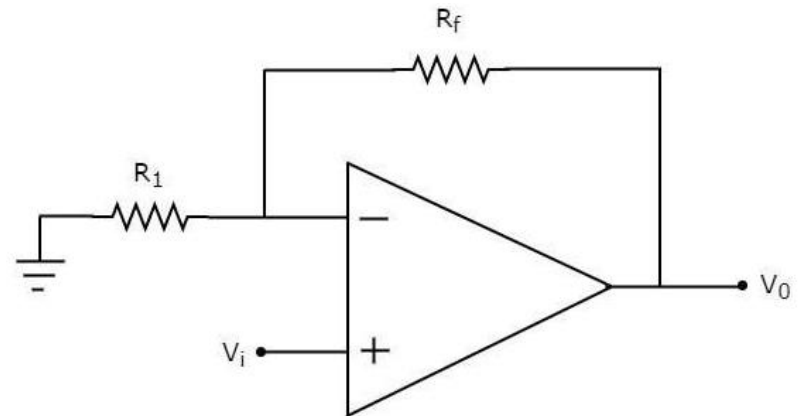
1. Inverting $\rightarrow V_o = -K V_i$
2. Non-inverting $\rightarrow V_o = +K V_i$
3. Buffer $\rightarrow V_o = +V_i$

Overview

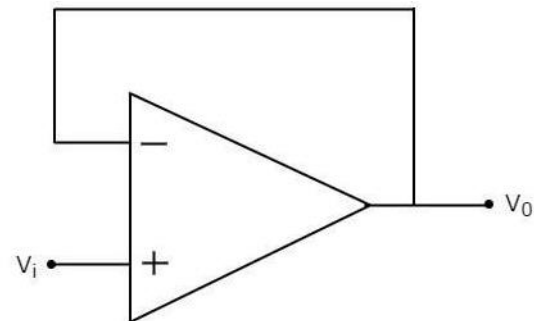
Inverting:



Non-inverting:

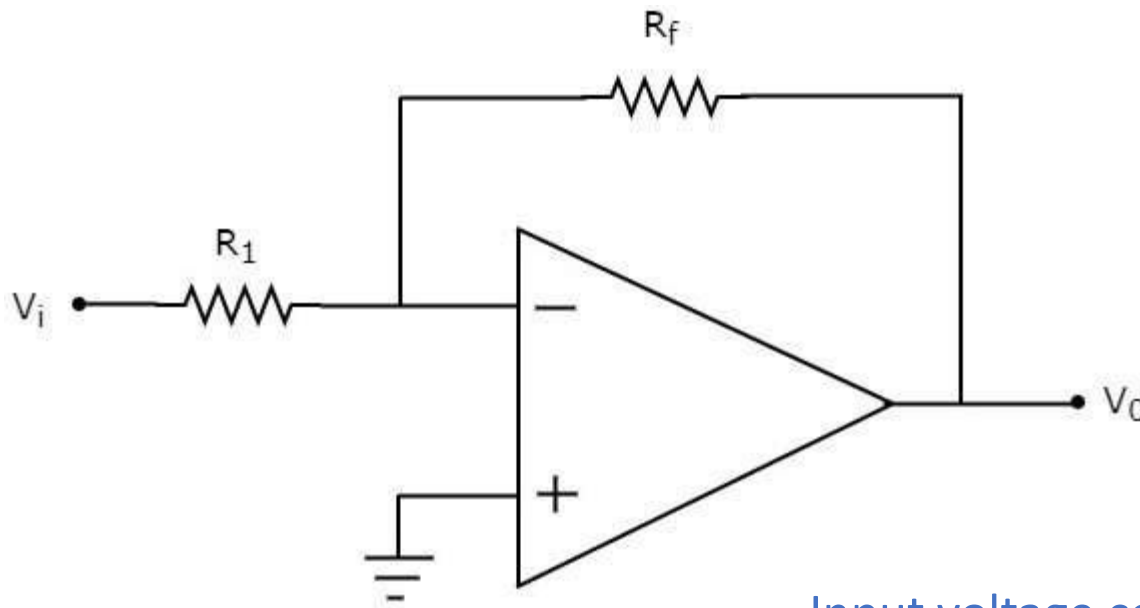


Buffer
(voltage follower):



Inverting amplifiers

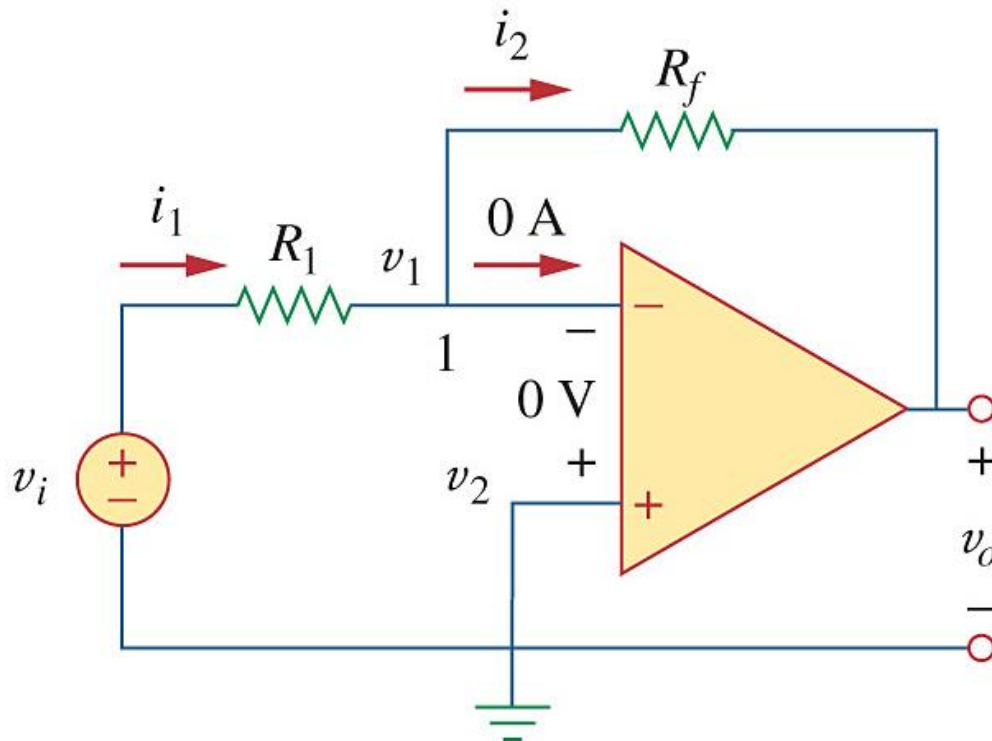
Reverses polarity of input signal while amplifying it



$$v_o = -\frac{R_f}{R_1} v_i$$

Input voltage connected to
inverting input
→ inverting amplifier

Inverting amplifiers



$$A_v = \frac{v_o}{v_i} = -\frac{R_f}{R_1}$$

$$\begin{aligned} i_1 &= i_2 = 0 \\ v_1 &= v_2 \end{aligned}$$

$$v_1 = v_2 = 0V$$

KCL at v_1 :

$$-i_1 + i_2 = 0$$

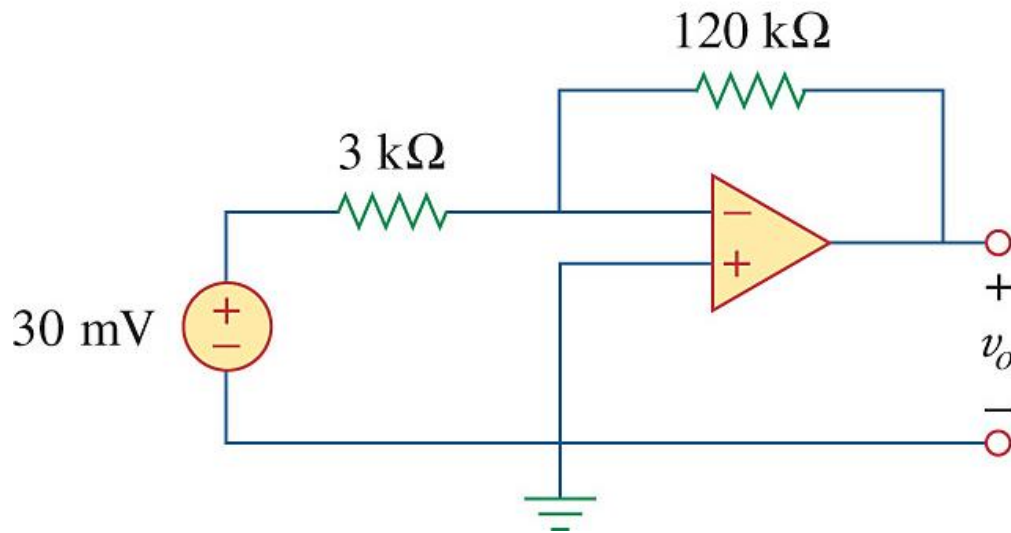
$$\frac{0 - v_i}{R_1} + \frac{0 - v_o}{R_f} = 0$$

$$\frac{v_o}{v_i} = -\frac{R_f}{R_1}$$

$$v_o = -\frac{R_f}{R_1} v_i$$

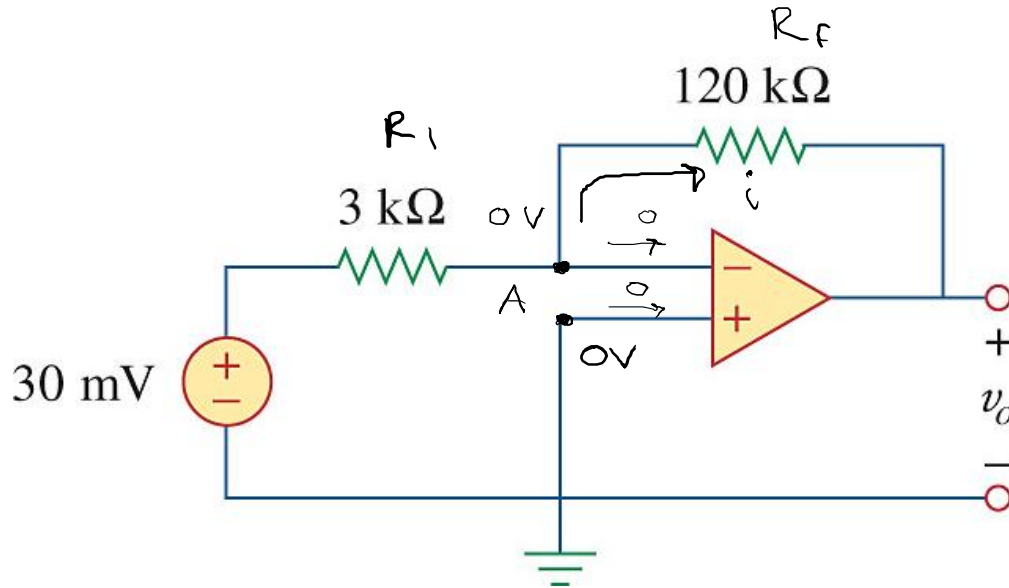
Example 1: Inverting amplifier

Determine the output voltage and current through the feedback resistor.



Example 1: Inverting amplifier

Determine the output voltage and current through the feedback resistor.



KCL at A:

$$\frac{0 - 30 \text{ mV}}{3 \text{ k}\Omega} + \frac{0 - V_o}{120 \text{ k}\Omega} = 0$$

$$\therefore i = \frac{-V_o}{120 \text{ k}\Omega} = \frac{-(-1.2 \text{ V})}{120 \text{ k}\Omega} = 10 \mu\text{A}$$

Inverting:

$$V_o = -\frac{R_f}{R_i} V_i$$

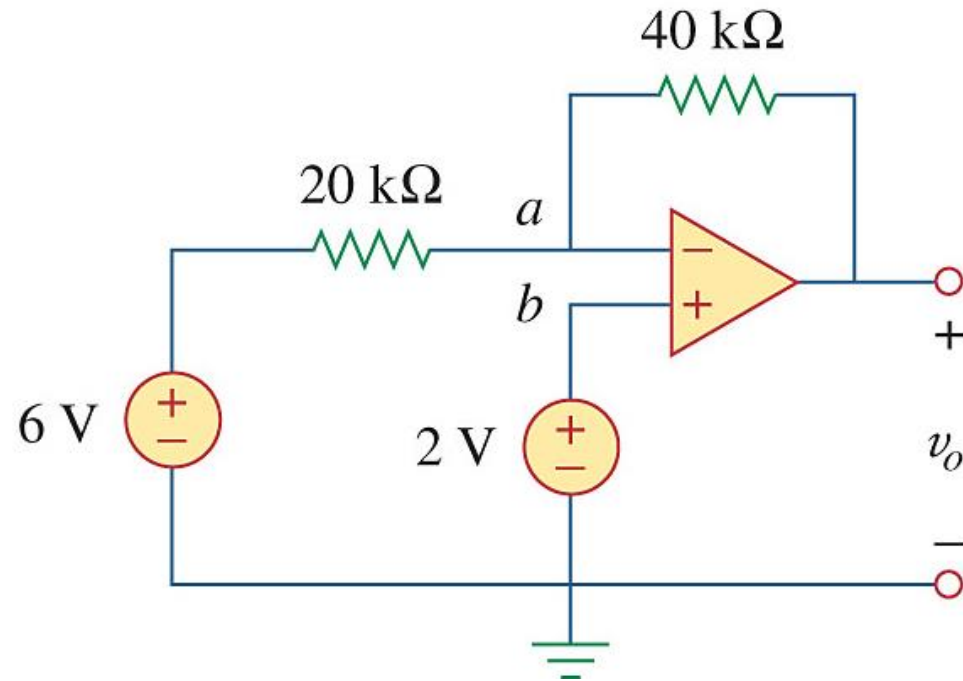
$$V_o = -\frac{120 \text{ k}}{3 \text{ k}} (30 \text{ mV})$$

$$V_o = -40 (30 \text{ mV})$$

$$V_o = -1.2 \text{ V}$$

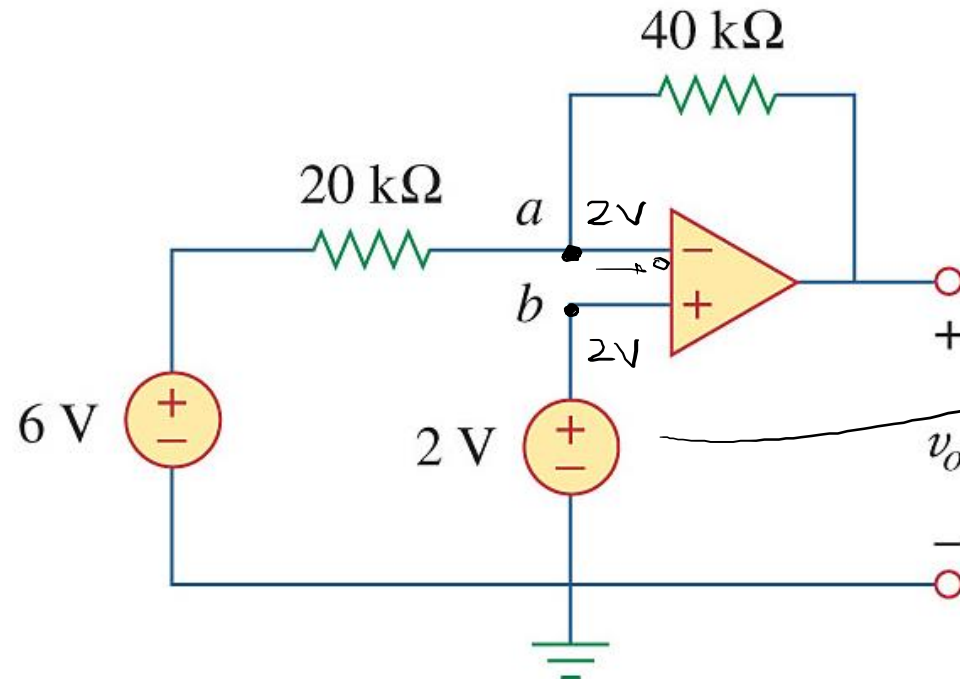
Example 2: Inverting amplifier

Determine v_o



Example 2: Inverting amplifier

Determine v_o



$\otimes$ KCL at a :

$$\frac{2 - 6}{20k} + \frac{2 - v_o}{40k} = 0$$

($\times 40k$):

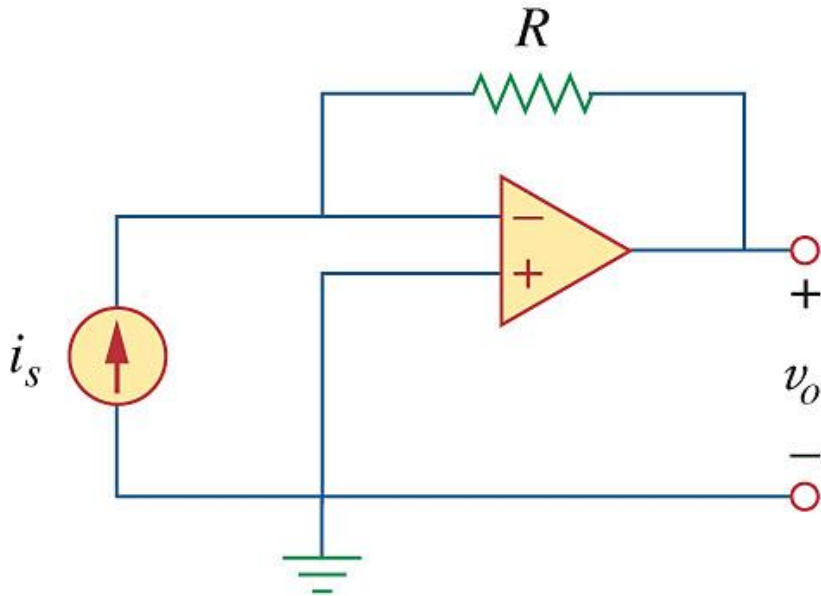
$$-8 + 2 - v_o = 0$$

not grounded,
so can't use
 $-\frac{R_f}{R_i}$ directly!

$$\therefore v_o = -6V$$

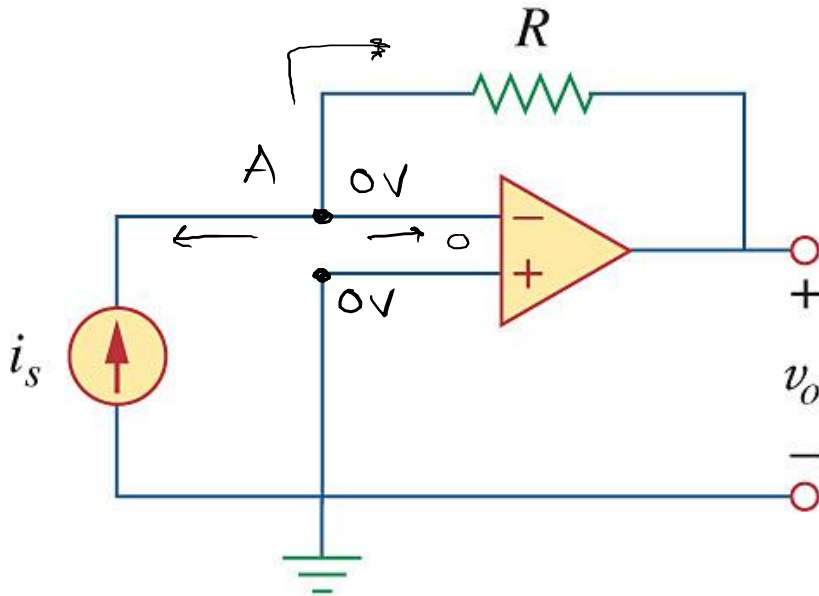
Example 3: Inverting amplifier

Special Case: Current-to-Voltage converter



Example 3: Inverting amplifier

Special Case: Current-to-Voltage converter



KCL at A:

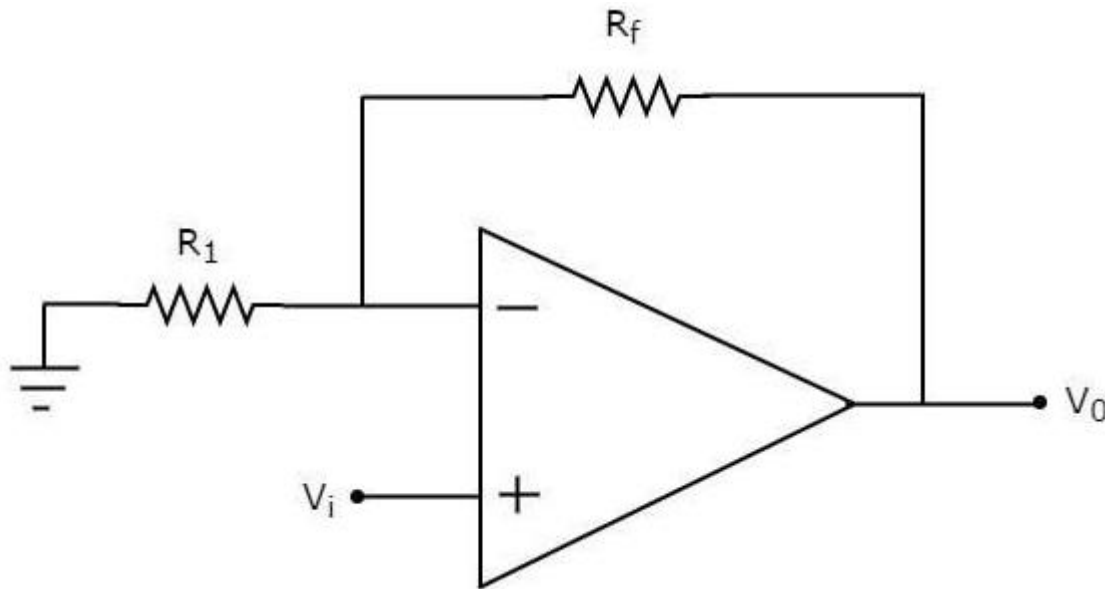
$$\frac{0 - v_o}{R} + -i_s = 0$$

$$i_s = -\frac{v_o}{R}$$

$$v_o = -i_s R \rightarrow$$

Non-inverting amplifiers

Designed to produce positive voltage gain



$$v_o = \left(1 + \frac{R_f}{R_1} \right) v_i$$

Input voltage connected to
non-inverting input
→ non-inverting amplifier

Non-inverting amplifiers

$$i_1 = i_2 = 0$$

$$v_1 = v_2$$

$$v_1 = v_2 = 0V$$

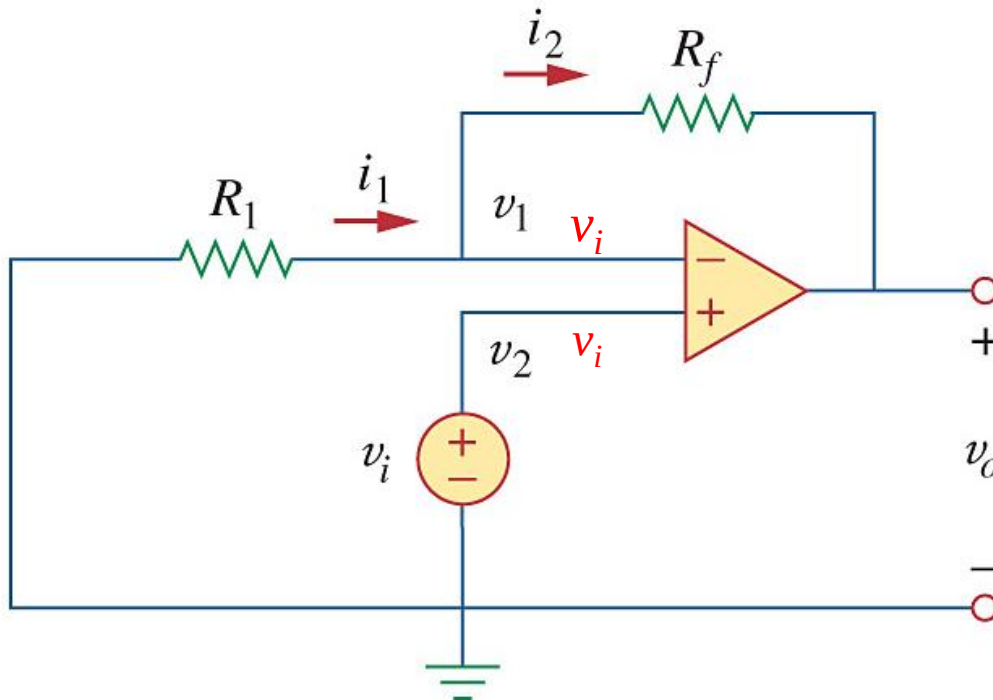
KCL at v_1 :

$$\frac{v_i - 0}{R_1} + \frac{v_i - v_o}{R_f} = 0$$

$$\frac{v_i - v_o}{v_i} = -\frac{R_f}{R_1}$$

$$1 - \frac{v_o}{v_i} = -\frac{R_f}{R_1}$$

$$\frac{v_o}{v_i} = 1 + \frac{R_f}{R_1}$$

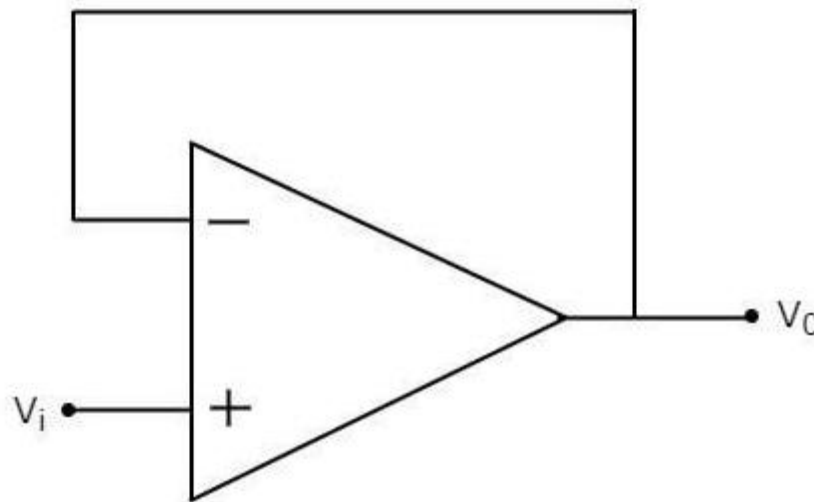


$$A_v = \frac{v_o}{v_i} = \left(1 + \frac{R_f}{R_1}\right)$$

$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Buffer or voltage follower

Special case of a non-inverting amplifier

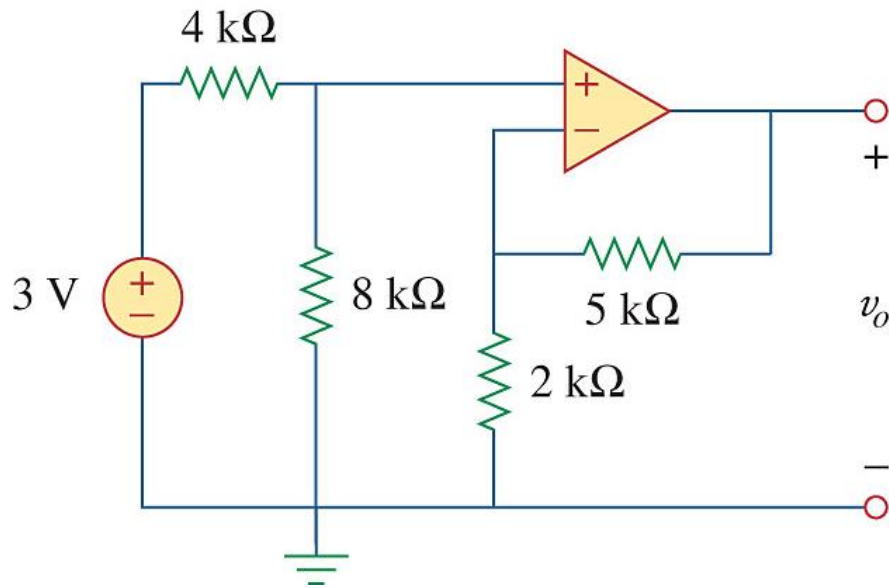


$$v_o = v_i$$

- Output = input
- Unity gain amplifier
- Used a **buffer** to isolate one stage of a circuit from another

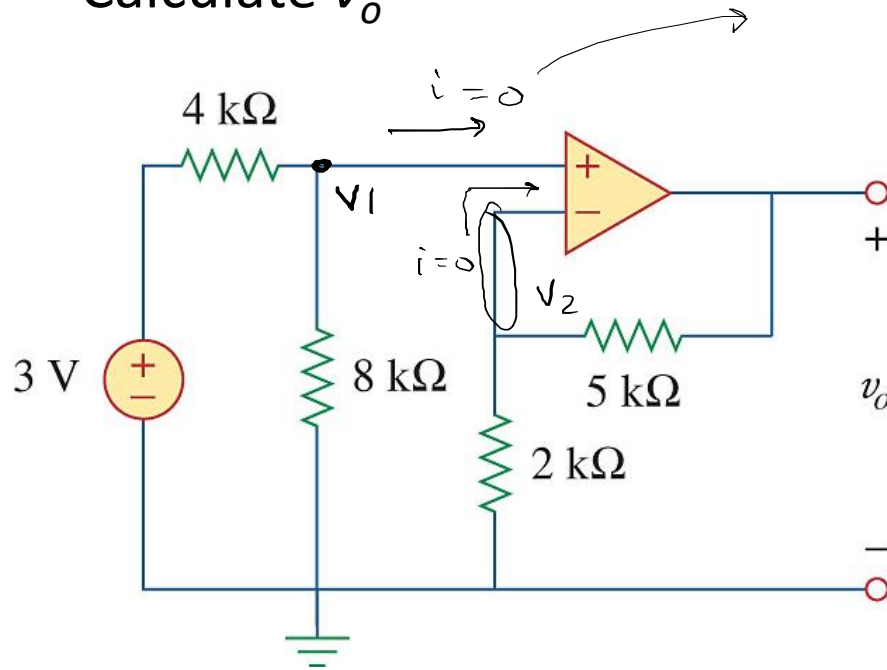
Example 1: Non-inverting amplifier

Calculate v_o



Example 1: Non-inverting amplifier

Calculate v_o



$\therefore$ Voltage division to get V_1 :

$$V_1 = 3V \left(\frac{8k}{8k + 4k} \right)$$

$$V_1 = 2V$$

KCL at V_2 :

$$\frac{V_2 - v_o}{5k} + \frac{V_2 - 0}{2k} = 0$$

and $V_2 = V_1 = 2V$:

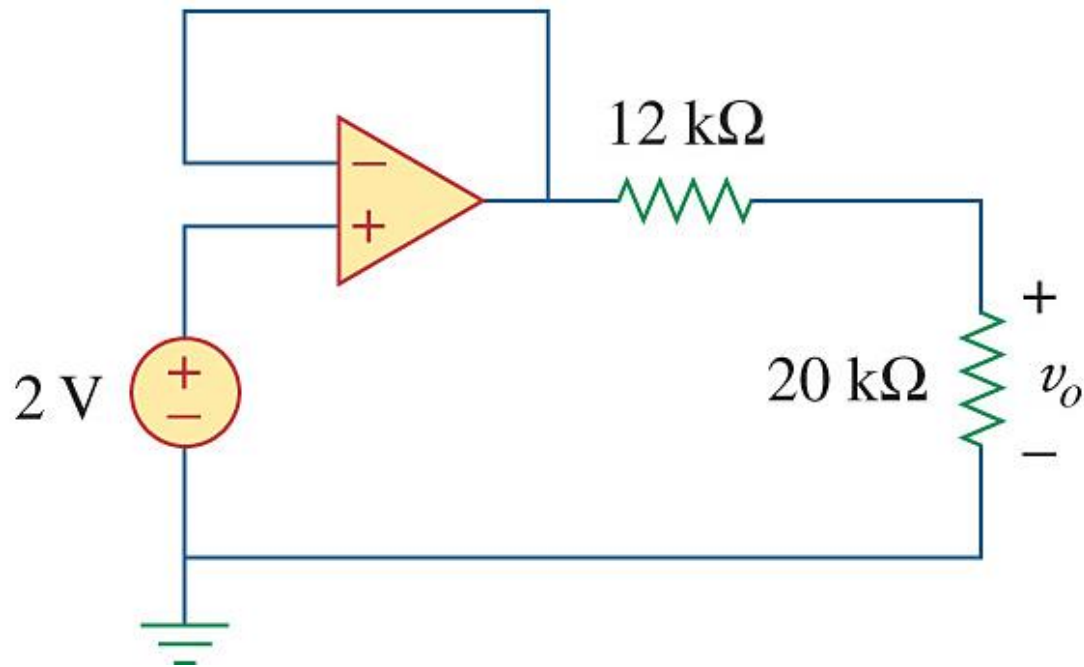
$$\therefore \frac{2 - v_o}{5k} + \frac{2}{2k} = 0$$

$$(\times 10k): \quad 4 - 2v_o + 10 = 0$$

$$\therefore \underline{v_o = 7V}$$

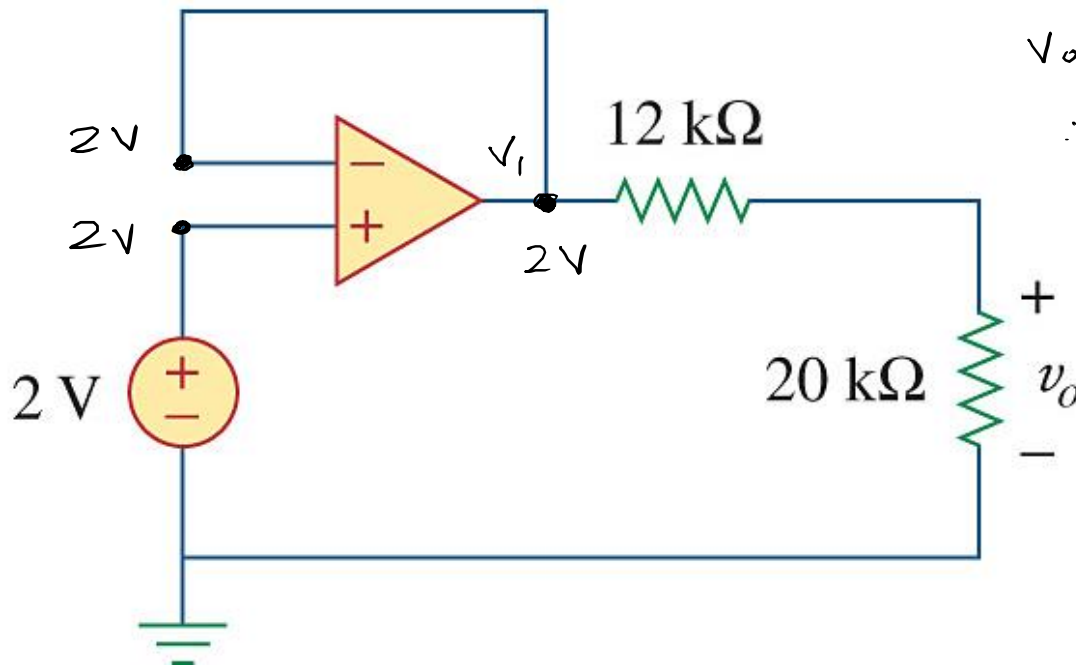
Example 2: Non-inverting amplifier

Calculate v_o



Example 2: Non-inverting amplifier

Calculate v_o



⊛ voltage follower / buffer:

$$V_o = V_i$$

$$\therefore V_i = 2V$$

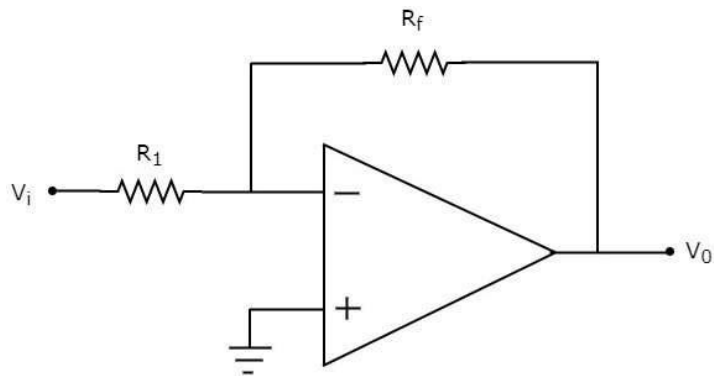
and V_o = voltage division:

$$V_o = 2V \left(\frac{20k}{20k + 12k} \right)$$

$$V_o = 1.25V$$

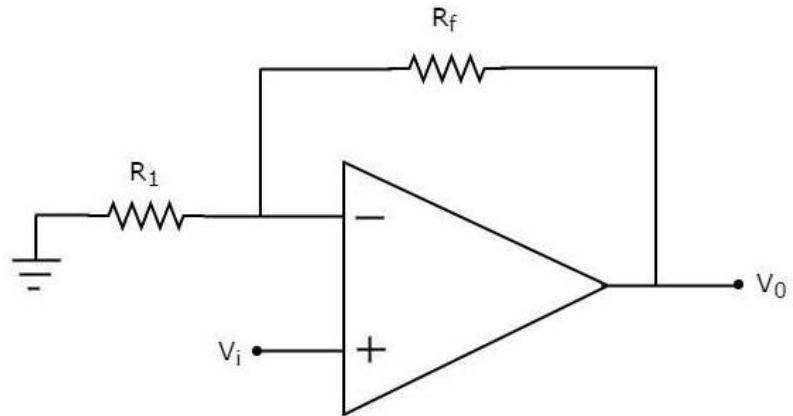
Summary:

Inverting:



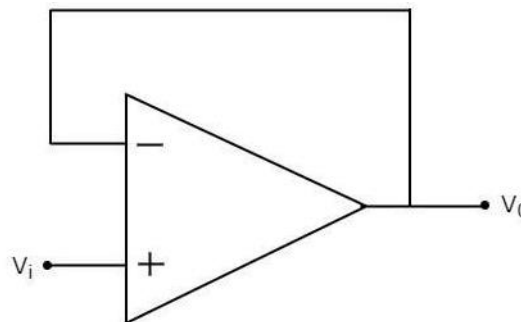
$$v_o = -\frac{R_f}{R_1} v_i$$

Non-inverting:



$$v_o = \left(1 + \frac{R_f}{R_1}\right) v_i$$

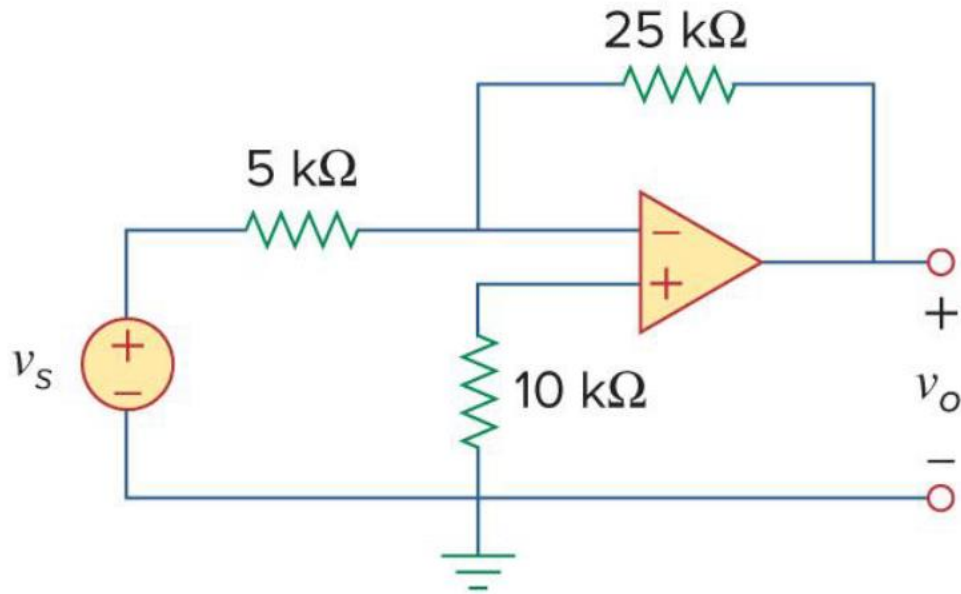
Buffer
(voltage follower):



$$v_o = v_i$$

Example 4: Inverting and Non-inverting amplifier circuits

- Determine v_o .



inverting op-amp configuration:

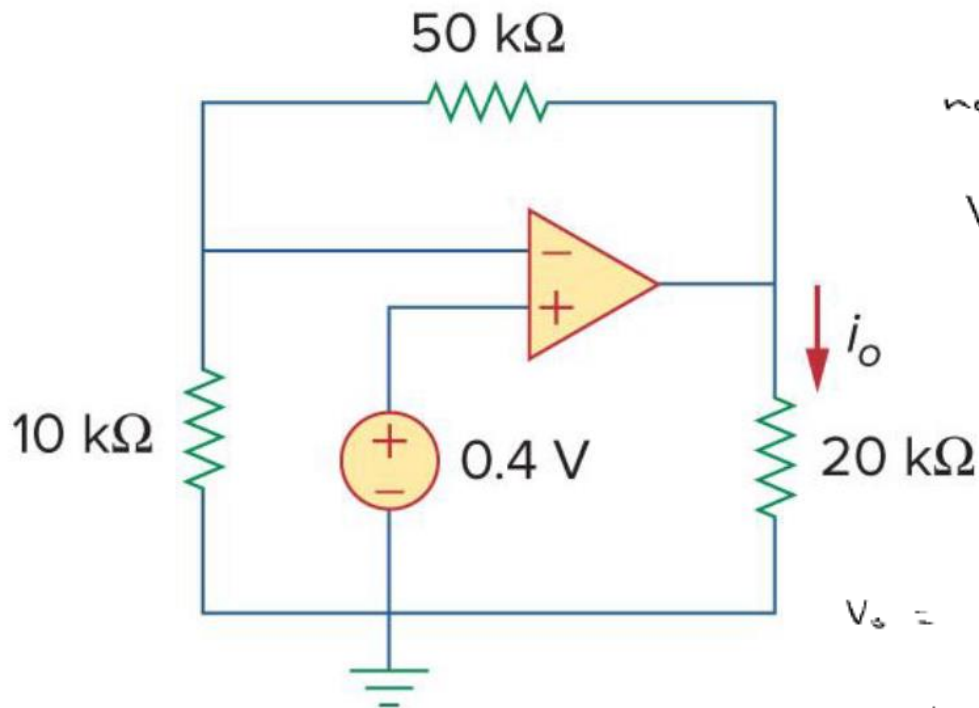
$$v_o = -\frac{R_f}{R_i} v_s = -\frac{25\text{ k}}{5\text{ k}} v_s$$

$$\therefore v_o = -5 v_s$$

→

Example 5: Inverting and Non-inverting amplifier circuits

- Determine i_o .



non-inverting op-amp configuration:

$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_s$$

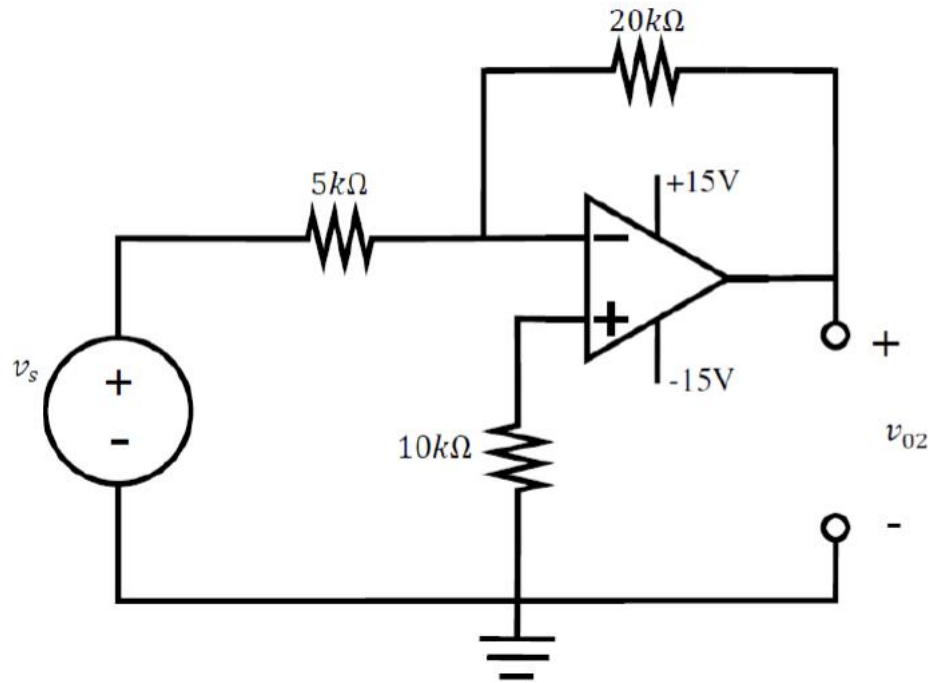
$$V_o = \left(1 + \frac{50k}{10k} \right) V_s$$

$$V_o = 6 (V_s) = 6 (0.4)$$

$$\therefore V_o = 2.4 \text{ V}, \quad i_o = \frac{V_o}{20k}$$
$$= 0.12 \text{ mA}$$

Example 6: Inverting and Non-inverting amplifier circuits

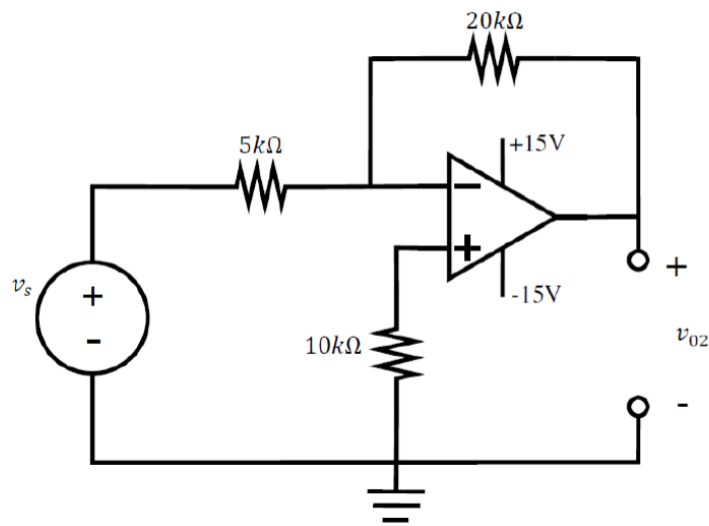
- a) Determine v_{o2} if $v_s = -1$ V.
- b) What is the minimum value v_s can be before the op amp goes into saturation?



inverting op-amp configuration:

$$v_{o2} = -\frac{R_F}{R_1} v_s = -\frac{20\text{k}}{5\text{k}} v_s \quad \text{① } v_s = -1\text{V}$$

$$\therefore v_{o2} = \underline{4\text{V}}$$



b) using ①: $\frac{V_{o2}}{20k} = -\frac{V_s}{5k} \quad \therefore V_o = -4V_s$

and V_o restricted by supply rails:

$$-15V \leq V_o \leq 15V$$

$$\therefore -15V \leq -4V_s \leq 15V$$

$$\therefore -3.75V \leq V_s \leq 3.75V \quad \therefore \min V_s = \underline{\underline{-3.75V}}$$